Labeling Learner States in Programming Sessions: A Reliability Study of Thirty-Second Windows

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Abstract

Studies that label novice programmers' behavioral states from logs choose a window length with little justification. We asked three trained raters to label the same 400 minutes of replayed programming sessions at three window lengths (10, 30 and 60 seconds) using a five-state scheme. Agreement was highest for 30-second windows (Fleiss' kappa 0.66), lower for 60 seconds (0.61), where windows often spanned two states, and lowest for 10 seconds (0.48), where raters lacked context. Most disagreements fell between "tinkering" and "stuck". We release our rater training materials and recommend 30-second windows with a one-window context margin.

1 Motivation

Window length is a hidden parameter in behavioral labeling. Short windows give fine temporal resolution but little context; long windows give context but blur transitions. Prior work has used anything from 5-second slices to whole-problem episodes. We isolate the choice and measure what it does to agreement.

2 Method

We sampled 40 ten-minute segments from 31 students in a Python CS1 course. Three raters (two TAs and a graduate student) labeled every segment three times, once per window length, with at least a week between passes and a randomized segment order. The scheme had five states: productive work, tinkering (rapid small edits and runs without a clear plan), stuck, help-seeking, and off-task.

Raters saw the replayed editor, the run history and any assistant prompts, plus 30 seconds of preceding context in every condition. Training took 90 minutes and used 12 practice segments with a reference key.

3 Results

Window	Windows labeled	Fleiss' kappa	% windows with 2+ states (rater flag)
10 s	2,400	0.48	3%
30 s	800	0.66	11%
60 s	400	0.61	27%

Disagreement concentrated on tinkering vs stuck (44% of all disagreements at 30 s). Raters who called a window "stuck" tended to weight the time since the last successful run; those who called it "tinkering" weighted the number of edits. Adding an explicit rule ("if no run succeeded in the previous 3 minutes and edits are reverted, label stuck") raised kappa on a held-out set from 0.66 to 0.70.

Help-seeking was the easiest state (pairwise agreement 0.91) because prompts and forum visits are visible events. Off-task was rarely labeled (2% of windows) and noisy.

4 Recommendations

• Use 30-second windows with at least 30 seconds of visible context before each window.

• Write decision rules for the tinkering/stuck boundary; most of the reliability gain comes from there.

• Report agreement per state, not only an overall kappa.

• Let raters flag mixed windows instead of forcing a single label; 11% is a real share of the data.

5 Limitations

Three raters from one institution; a five-state scheme. Agreement may differ with other schemes or with raters who did not teach the course.

References

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